

LECTURE: SEPARATION TECHNIQUES IN BIOCHEMISTRY

Course Information

- Course: Advanced Biochemistry / Chemical Pathology
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- Department: Medical Biochemistry, Rivers State University
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LEARNING OBJECTIVES

Upon completion of this lecture, students should be able to:

1. Explain the fundamental principles governing biomolecule separation
2. Select appropriate separation techniques based on physicochemical properties
3. Describe the mechanisms, advantages, and limitations of each method
4. Interpret chromatographic and electrophoretic data
5. Apply separation techniques in clinical diagnostic and research contexts
6. Troubleshoot common problems in separation procedures

SECTION 1: INTRODUCTION AND FUNDAMENTAL PRINCIPLES

1.1 Why Separation is Essential in Biochemistry

Biological samples are extraordinarily complex mixtures containing:

Table

Component Class	Examples	Approximate Count in Human Plasma
Proteins	Albumin, immunoglobulins, enzymes	~10,000 distinct species

Lipids	Cholesterol, triglycerides, phospholipids	~1,000 lipid species
Nucleic acids	DNA, RNA, miRNA	Multiple forms
Carbohydrates	Glucose, glycoproteins, glycolipids	Hundreds of glycoforms
Small molecules	Metabolites, hormones, vitamins	~5,000+ metabolites
Ions	Na ⁺ , K ⁺ , Ca ²⁺ , Cl ⁻	Essential electrolytes

Key Principle: No single analytical technique can identify or quantify a target molecule in such complexity without prior separation.

1.2 Physicochemical Properties Exploited for Separation

Table

Property	Separation Technique
Size/Molecular Weight	Dialysis, ultrafiltration, size-exclusion chromatography, gel electrophoresis
Charge	Ion-exchange chromatography, electrophoresis, isoelectric focusing
Hydrophobicity	Hydrophobic interaction chromatography, reversed-phase HPLC

Solubility	Salting out, organic solvent precipitation, crystallization
Affinity	Affinity chromatography, immunoprecipitation
Density	Density gradient centrifugation, ultracentrifugation
Volatility	Distillation, gas chromatography
Diffusion Coefficient	Dialysis, ultrafiltration
Biological Activity	Bioassay-guided fractionation

1.3 Resolution: The Critical Parameter

Resolution (R_s) is defined as:

$$R_s$$

$$s$$

$$=$$

$$W$$

$$1$$

$$+W$$

$$2(t$$

$$R2$$

$$-t$$

$$R1$$

)

Where:

- t_{r1}, t_{r2} = retention times of two adjacent peaks
- W_1, W_2 = peak widths at baseline

Table

Resolution Value	Separation Quality
$R_s < 1.0$	Overlapping peaks; inadequate separation
$R_s = 1.0$	~4% overlap; marginal separation
$R_s = 1.5$	Baseline separation (6σ); adequate for quantification
$R_s > 2.0$	Complete separation; optimal for preparative work

SECTION 2: CENTRIFUGATION TECHNIQUES

2.1 Principle

Centrifugation separates particles based on size, shape, and density using centrifugal force:

$$F = m \cdot \omega$$

2

· r

Where:

- F = centrifugal force
- m = mass of particle
- ω = angular velocity (rad/s)
- r = radial distance from axis

Relative Centrifugal Force (RCF or g-force):

$$RCF = 1.118 \times 10$$

−5

$\times r \times (rpm)$

2

2.2 Types of Centrifugation

Table

Type	RCF Range	Application	Key Principle
Differential Centrifugation	500–10,000 × g	Cell organelle isolation	Sequential pelleting by increasing speed
Density Gradient Centrifugation	10,000–100,000 × g	Subcellular fractionation, lipoproteins	Buoyant density equilibrium
Rate-Zonal Centrifugation	50,000–200,000 × g	RNA separation, ribosomes	Sedimentation rate in sucrose gradient
Isopycnic (Equilibrium) Centrifugation	100,000–400,000 × g	Plasmid DNA, viral particles	Density matching with CsCl
Ultracentrifugation	>100,000 × g	Protein complexes, membrane vesicles	High-speed sedimentation
Analytical Ultracentrifugation	Variable	Molecular weight determination, sedimentation coefficient	Real-time optical detection

2.3 Clinical Application: Lipoprotein Separation

Ultracentrifugation (Havel Method):

- VLDL: $d < 1.006$ g/mL (floats at density 1.006)
- LDL: $d = 1.006\text{--}1.063$ g/mL
- HDL: $d = 1.063\text{--}1.21$ g/mL

Correlation with Cardiovascular Risk:

Table

Lipoprotein	Density (g/mL)	Clinical Significance
Chylomicrons	<0.95	Postprandial; pancreatitis risk if elevated
VLDL	$0.95\text{--}1.006$	Triglyceride transport; precursor to LDL
LDL	$1.006\text{--}1.063$	"Bad cholesterol"; atherosclerotic risk
HDL	$1.063\text{--}1.21$	"Good cholesterol"; reverse cholesterol transport

SECTION 3: PRECIPITATION TECHNIQUES

3.1 Salting Out (Ammonium Sulfate Fractionation)

Principle: High salt concentrations reduce protein solubility by competing for water molecules (hydration shell disruption).

Mechanism:

- At low ionic strength: "Salting in" — increased solubility
- At high ionic strength: "Salting out" — decreased solubility

Ammonium Sulfate Saturation Table:

Table

% Saturation	Proteins Precipitated	Typical Use
0–20%	Fibrinogen	Removal of bulk contaminants
20–40%	Globulins (some)	Partial purification
40–60%	Most globulins	IgG enrichment
60–80%	Albumin, most enzymes	General protein precipitation
80–100%	Highly soluble proteins	Complete precipitation

Advantages: Inexpensive, preserves enzyme activity, scalable

Limitations: Low resolution, co-precipitation of contaminants

3.2 Organic Solvent Precipitation

Table

Solvent	Mechanism	Typical Concentration
Acetone	Reduces dielectric constant	3–4 volumes
Ethanol	Dehydration, reduces dielectric constant	2–3 volumes

Trichloroacetic acid (TCA)	Denaturation, pH disruption	5–10% final
Perchloric acid (PCA)	Denaturation, precipitation of proteins	0.2–0.6 M

Clinical Application: Protein precipitation in drug analysis (removes plasma proteins before HPLC analysis of small molecule drugs).

3.3 Isoelectric Precipitation

Principle: Proteins have minimum solubility at their isoelectric point (pI) where net charge = 0.

$$pI =$$

2

$$pK$$

$$\alpha 1$$

$$+pK$$

$$\alpha 2$$

(for simple amino acids)

Example: Casein precipitation in cheese making (pI ~4.6)

SECTION 4: DIALYSIS AND ULTRAFILTRATION

4.1 Dialysis

Principle: Separation based on molecular size using semipermeable membranes.

Mechanism: Small molecules (<MWCO) diffuse across membrane; macromolecules are retained.

Key Parameters:

Table		
Parameter	Description	Typical Range
MWCO (Molecular Weight Cut-Off)	Pore size specification	1–1,000 kDa
Membrane Material	Regenerated cellulose, PVDF, PES	Various
Dialysis Time	Depends on concentration gradient	4–24 hours
Buffer Volume	Typically 100–500× sample volume	–

Clinical Application: Hemodialysis — removal of urea, creatinine, and toxins from blood in renal failure patients.

4.2 Ultrafiltration

Principle: Pressure-driven separation across semipermeable membranes.

Types:

Table	
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Type	Pressure Source	Application
Dead-end ultrafiltration	Centrifugal or gas pressure	Concentration of protein samples
Cross-flow ultrafiltration	Pump-driven tangential flow	Large-scale purification, virus removal
Stirred-cell ultrafiltration	Gas pressure with stirring	Concentration and diafiltration

Diafiltration: Simultaneous buffer exchange and concentration — critical for removing small molecule contaminants (e.g., salts, detergents) from protein preparations.

SECTION 5: CHROMATOGRAPHY — FUNDAMENTALS

5.1 General Principles

Chromatography separates components based on differential partitioning between:

- Stationary phase (solid or liquid supported on solid)
- Mobile phase (liquid or gas)

Retention Factor (k'):

k

,

=

t

0

t

R

$-t$

0

=

time in mobile phase

time in stationary phase

Selectivity Factor (α):

$\alpha =$

k

1

,

k

2

,

5.2 Chromatographic Performance Parameters

Table

Parameter	Symbol	Definition	Optimal Value
Retention time	t_r	Time from injection to peak maximum	Analyte-specific
Dead time	t_0	Time for unretained compound	—
Peak width	W	Width at baseline (4σ)	Narrow
Theoretical plates	N	$N = 16(t_r/W)^2$	High (>10,000)
Plate height	H	$H = L/N$ (L = column length)	Low
Peak asymmetry	A_s	Ratio of trailing to leading half-width	0.9–1.2

Van Deemter Equation:

$$H = A +$$

$$B$$

$$C \cdot u$$

$$+ C \cdot u$$

Where:

- A = eddy diffusion (multiple paths)
- B = longitudinal diffusion
- C = resistance to mass transfer
- u = linear velocity of mobile phase

SECTION 6: LIQUID CHROMATOGRAPHY TECHNIQUES

6.1 Size-Exclusion Chromatography (SEC / Gel Filtration)

Principle: Separation by molecular size — larger molecules elute first (excluded from pores), smaller molecules enter pores and elute later.

Stationary Phase: Porous beads (cross-linked dextran, agarose, polyacrylamide)

Common Media:

Table

Medium	Fractionation Range (Proteins)	Application
Sephadex G-25	1–5 kDa	Desalting, buffer exchange
Sephadex G-75	3–70 kDa	Peptide separation
Sephadex G-100	4–150 kDa	Small protein purification
Sephadex G-200	5–600 kDa	Large proteins, antibodies
Sepharose CL-6B	10–4,000 kDa	Polysaccharides, complexes

Superdex 200	10–600 kDa	High-resolution protein analysis
Superose 6	5–5,000 kDa	Very large complexes, viruses

Clinical Application: Detection of Bence Jones proteins (free light chains) in urine – diagnostic for multiple myeloma.

6.2 Ion-Exchange Chromatography (IEC)

Principle: Separation based on charge – reversible binding to charged stationary phase.

Types:

Table

Type	Stationary Phase Charge	Binds	Elution Method
Cation exchange (CM, SP, S)	Negative (carboxymethyl, sulfopropyl)	Positively charged proteins ($pH < pI$)	Increasing salt or pH
Anion exchange (DEAE, Q, QAE)	Positive (diethylaminoethyl, quaternary ammonium)	Negatively charged proteins ($pH > pI$)	Increasing salt or decreasing pH

Elution Strategies:

1. Gradient elution: Continuous increase in salt concentration (NaCl)
2. Step elution: Discrete salt steps
3. pH elution: Changing pH to alter protein charge

Clinical Application: Separation of hemoglobin variants (HbA, HbS, HbC) for sickle cell and thalassemia screening.

6.3 Hydrophobic Interaction Chromatography (HIC)

Principle: Separation based on surface hydrophobicity – proteins bind to hydrophobic ligands at high salt; elute with decreasing salt.

Ligands: Phenyl, butyl, octyl groups

Mechanism: High salt promotes hydrophobic interactions (salting-out effect); low salt disrupts them.

Advantage: Preserves native protein structure (non-denaturing) – ideal for enzyme purification.

6.4 Affinity Chromatography

Principle: Biospecific recognition – target molecule binds to immobilized ligand with high specificity.

Common Ligand-Target Pairs:

Table		
Ligand	Target Molecule	Application
Protein A / Protein G	IgG antibodies	Antibody purification
Lectins (ConA, WGA)	Glycoproteins	Glycoprotein isolation

Heparin	Coagulation factors, lipases	Blood factor purification
NAD ⁺ / NADP ⁺	Dehydrogenases	Enzyme purification
Calmodulin	Calmodulin-binding proteins	Signal transduction research
Boronate	cis-Diol groups (glycoproteins)	Glycoprotein enrichment
Immobilized metal (IMAC)	His-tagged proteins	Recombinant protein purification

Clinical Application: Immunoaffinity chromatography for therapeutic drug monitoring (e.g., monoclonal antibody quantification).

6.5 Reversed-Phase High-Performance Liquid Chromatography (RP-HPLC)

Principle: Separation based on hydrophobicity – nonpolar stationary phase, polar mobile phase.

Stationary Phase: C18 (octadecylsilane), C8, C4 bonded silica

Mobile Phase: Water/acetonitrile or water/methanol with ion-pairing agents (TFA)

Elution: Increasing organic solvent gradient

Clinical Applications in Chemical Pathology:

Table

Analyte	Clinical Significance	Typical Method
HbA1c	Diabetes diagnosis and monitoring	HPLC (NGSP certified)
Amino acids	Inborn errors of metabolism	RP-HPLC with derivatization
Therapeutic drugs	Drug level monitoring	RP-HPLC with UV or MS detection
Vitamins (D, B12, folate)	Nutritional deficiency assessment	RP-HPLC or LC-MS/MS
Steroid hormones	Endocrine disorders	RP-HPLC or LC-MS/MS
Bile acids	Liver function, cholestasis	RP-HPLC
Catecholamines	Pheochromocytoma diagnosis	RP-HPLC with electrochemical detection

HbA1c by HPLC (Clinical Standard):

- HbA1c elutes at characteristic retention time
- Separation from HbF, HbS, HbC variants
- NGSP-certified methods required for clinical reporting

SECTION 7: GAS CHROMATOGRAPHY (GC)

7.1 Principle

Separation of volatile or volatilizable compounds using:

- Stationary phase: Liquid or polymer coated on capillary column
- Mobile phase: Inert gas (helium, nitrogen, hydrogen)

7.2 Clinical Applications

Table

Application	Analytes	Detection Method
Volatile organic compounds (VOCs)	Ethanol, methanol, isopropanol	FID (Flame Ionization)
Toxicology screening	Drugs of abuse, pesticides	MS (Mass Spectrometry)
Organic acid analysis	Urine organic acids (inborn errors)	MS
Fatty acid methyl esters (FAME)	Essential fatty acid profiles	FID or MS
Amino acid analysis	Derivatized amino acids	MS
Steroid profiling	Urine steroid metabolites	MS

GC-MS in Newborn Screening:

- Detection of inborn errors of metabolism
- Organic acidemias, fatty acid oxidation disorders
- High sensitivity and specificity

SECTION 8: ELECTROPHORESIS

8.1 Fundamental Principles

Electrophoretic Mobility (μ):

$$\mu =$$

$$E$$

$$v$$

$$=$$

$$6\pi\eta r$$

$$q$$

Where:

- v = velocity of migration
- E = electric field strength
- q = net charge
- η = viscosity
- r = Stokes radius

Key Principle: Molecules migrate at rates proportional to their charge-to-size ratio.

8.2 Agarose Gel Electrophoresis

Principle: Separation of nucleic acids (DNA, RNA) by size.

Table

Parameter	Setting

Gel concentration	0.5–2.0% agarose
Buffer	TAE (Tris-acetate-EDTA) or TBE (Tris-borate-EDTA)
Voltage	5–10 V/cm
Detection	Ethidium bromide or SYBR Safe (fluorescence)

Resolution vs. Gel Concentration:

Table

Gel %	Optimal Separation Range	Application
0.5%	1–30 kb	Large DNA fragments, genomic DNA
0.8%	0.5–12 kb	Standard DNA electrophoresis
1.0%	0.3–10 kb	General purpose
1.5%	0.2–4 kb	PCR products
2.0%	0.1–3 kb	Small fragments, high resolution

Clinical Application: Serum protein electrophoresis (SPE) – diagnosis of monoclonal gammopathies (multiple myeloma).

8.3 Polyacrylamide Gel Electrophoresis (PAGE)

Principle: Higher resolution than agarose for proteins and small nucleic acids.

Types:

Table

Type	Denaturing?	Application
Native PAGE	No	Active enzyme separation, protein complexes
SDS-PAGE	Yes (SDS + reducing agent)	Molecular weight determination, subunit analysis
Urea-PAGE	Yes (urea)	DNA sequencing, single-strand conformation

SDS-PAGE Mechanism:

- SDS binds to proteins (1.4 g SDS/g protein)
- Confers uniform negative charge proportional to mass
- Separation purely by molecular weight

Laemmli Buffer System:

- Stacking gel (pH 6.8): Concentrates sample
- Separating gel (pH 8.8): Resolves by size

Protein Molecular Weight Markers:

Table

Protein	Molecular Weight (kDa)

Thyroglobulin	330
Ferritin	220
BSA (dimer)	132
BSA	66
Ovalbumin	45
Carbonic anhydrase	29
β -Lactoglobulin	18
Lysozyme	14
Aprotinin	6
Insulin B chain	3.4

Clinical Application: Western blotting – confirmatory test for HIV (detection of anti-HIV antibodies), Lyme disease, and autoimmune disorders.

8.4 Isoelectric Focusing (IEF)

Principle: Separation by isoelectric point (pI) in a pH gradient.

Mechanism: Proteins migrate to the position where pH = pI (net charge = 0).

Immobilized pH Gradient (IPG) strips:

- pH 3–10 (broad range)
- pH 4–7, 6–9 (narrow range, higher resolution)

Clinical Application: Hemoglobin electrophoresis — diagnosis of hemoglobinopathies:

- HbA (pI ~7.0)
- HbS (pI ~6.9, sickle cell)
- HbC (pI ~7.1)
- HbF (fetal hemoglobin)

8.5 Two-Dimensional Gel Electrophoresis (2D-PAGE)

First dimension: IEF (pI separation)

Second dimension: SDS-PAGE (molecular weight separation)

Resolution: Can resolve >10,000 protein spots from a complex mixture

Clinical Application: Proteomics biomarker discovery — differential protein expression in cancer vs. normal tissue.

SECTION 9: CAPILLARY ELECTROPHORESIS (CE)

9.1 Principle

Electrophoresis performed in narrow-bore fused silica capillaries (25–100 μm ID).

Advantages over conventional gel electrophoresis:

- High efficiency ($N > 100,000$ theoretical plates)
- Rapid separation (minutes)
- Minimal sample requirement (nanoliters)
- Automated quantification

Modes of CE:

Table

Mode	Separation Mechanism	Application
Capillary Zone Electrophoresis (CZE)	Charge-to-size ratio	Small ions, drugs, peptides
Capillary Gel Electrophoresis (CGE)	Size in sieving matrix	DNA sequencing, protein sizing
Capillary Isoelectric Focusing (CIEF)	pI	Protein isoform separation
Micellar Electrokinetic Chromatography (MEKC)	Partitioning into micelles	Neutral molecules, chiral separation
Capillary Electrochromatography (CEC)	Chromatographic + electrophoretic	Small molecules, peptides

9.2 Clinical Applications

Table

Application	Clinical Significance	Advantages over HPLC
Serum protein electrophoresis	Monoclonal gammopathy detection	Faster, higher resolution, automated

Hemoglobin analysis	Hemoglobinopathy screening	Rapid, quantitative, automated
Carbohydrate-deficient transferrin (CDT)	Alcohol abuse marker	High specificity
Therapeutic drug monitoring	Immunosuppressants, antibiotics	Minimal sample, fast
Urinary organic acids	Inborn errors of metabolism	Direct injection possible
Cerebrospinal fluid proteins	Multiple sclerosis, CNS disorders	High sensitivity

SECTION 10: ADVANCED SEPARATION TECHNIQUES

10.1 High-Performance Liquid Chromatography-Mass Spectrometry (LC-MS/MS)

Principle: Chromatographic separation coupled with tandem mass spectrometry for identification and quantification.

Clinical Applications:

Table

Application	Analytes	Clinical Impact
Newborn screening	Amino acids, acylcarnitines, steroids	Detection of >50 inborn errors
Therapeutic drug monitoring	Immunosuppressants (tacrolimus, cyclosporine), antiepileptics	Precision dosing, toxicity prevention
Vitamin D analysis	25-OH Vitamin D2 and D3	Accurate deficiency assessment
Steroid profiling	Cortisol, aldosterone, androgens	Endocrine disorder diagnosis
Protein quantification	Procalcitonin, troponin	High-sensitivity biomarker measurement
Toxicology	Drugs of abuse, pesticides	Definitive identification

Advantages: High sensitivity (pg/mL), high specificity (MS/MS transitions), multiplexing capability.

10.2 Field-Flow Fractionation (FFF)

Principle: Separation in an open channel by combined action of flow and external field (cross-flow, thermal, gravitational).

Application: Separation of macromolecules and nanoparticles too large for SEC:

- Lipoproteins
- Viruses
- Protein aggregates
- Nanoparticles

10.3 Counter-Current Chromatography (CCC)

Principle: Liquid-liquid partition without solid stationary phase — no irreversible adsorption.

Application: Purification of natural products from crude extracts:

- Polyphenols from plant extracts
- Alkaloids
- Antibiotics

SECTION 11: CLINICAL CASE STUDIES AND TROUBLESHOOTING

11.1 Case Study 1: Multiple Myeloma Diagnosis

Presentation: 65-year-old male with bone pain, anemia, elevated total protein.

Separation Workflow:

1. Serum protein electrophoresis (SPE): Monoclonal spike in γ -region
2. Immunofixation electrophoresis (IFE): Confirms IgG κ monoclonal protein
3. Serum free light chain assay: Elevated κ/λ ratio
4. Bone marrow biopsy: >10% plasma cells

Key Separation Technique: Agarose gel electrophoresis with densitometric scanning.

11.2 Case Study 2: Inborn Error of Metabolism

Presentation: Newborn with poor feeding, lethargy, metabolic acidosis.

Separation Workflow:

1. Blood spot amino acid analysis (LC-MS/MS): Elevated phenylalanine
2. Urine organic acids (GC-MS): Phenylpyruvic acid, phenyllactic acid
3. Confirmatory testing: Phenylalanine hydroxylase mutation analysis

Diagnosis: Phenylketonuria (PKU)

Key Separation Techniques: LC-MS/MS for amino acids, GC-MS for organic acids.

11.3 Troubleshooting Common Problems

Table

Problem	Possible Cause	Solution
Broad peaks in HPLC	Column degradation, extra-column band broadening	Replace column, check fittings
Peak tailing	Silanol interactions, column overload	Add ion-pairing agent, reduce sample load
No peaks in electrophoresis	Sample degradation, incorrect buffer	Check sample integrity, verify buffer pH
Smearing in SDS-PAGE	Insufficient reduction, sample overload	Increase reducing agent, dilute sample
Baseline drift in GC	Column bleed, temperature instability	Condition column, check oven

Poor resolution in IEC	Incorrect pH, too high flow rate	Optimize pH, reduce flow rate
Protein precipitation during HIC	Too high salt, incompatible buffer	Reduce initial salt, check buffer compatibility

SECTION 12: SUMMARY AND COMPARATIVE OVERVIEW

12.1 Technique Selection Guide

Table

Target Property	Primary Technique	Secondary Technique
Molecular weight	SEC, SDS-PAGE	MALDI-TOF MS, analytical ultracentrifugation
Charge	IEC, IEF, CZE	2D-PAGE
Hydrophobicity	RP-HPLC, HIC	HIC (native conditions)
Affinity/Biospecificity	Affinity chromatography	Immunoprecipitation, pull-down assays
Volatility	GC, GC-MS	Headspace analysis

Complex mixtures	LC-MS/MS	2D-LC, multidimensional chromatography
Live cells/organelles	Density gradient centrifugation	Flow cytometry

12.2 Key Performance Comparison

Table					
Technique	Resolution	Sensitivity	Throughput	Cost	Skill Level
Centrifugation	Low	Moderate	High	Low	Low
Precipitation	Low	Moderate	High	Low	Low
Dialysis/UF	Low	Moderate	Moderate	Low	Low
SEC	Moderate	Moderate	Moderate	Moderate	Moderate
IEC	High	High	High	Moderate	Moderate

RP-HPLC	High	High	High	Moderate-High	Moderate
Affinity chromatography	Very High	Very High	Moderate	High	High
GC	High	High	High	Moderate	Moderate
SDS-PAGE	Moderate	Moderate	Moderate	Low	Low
2D-PAGE	Very High	Moderate	Low	Moderate	High
Capillary electrophoresis	Very High	High	High	Moderate-High	High
LC-MS/MS	Very High	Very High	High	Very High	Very High

RECOMMENDED READINGS

1. Wilson, K., & Walker, J. (2018). *Principles and Techniques of Biochemistry and Molecular Biology* (8th ed.). Cambridge University Press.
2. Berg, J. M., Tymoczko, J. L., & Stryer, L. (2019). *Biochemistry* (9th ed.). W.H. Freeman. (Chapter 3: Exploring Proteins)

3. Scopes, R. K. (2013). *Protein Purification: Principles and Practice* (3rd ed.). Springer.
4. Snyder, L. R., Kirkland, J. J., & Dolan, J. W. (2011). *Introduction to Modern Liquid Chromatography* (3rd ed.). Wiley.
5. Westermeier, R. (2016). *Electrophoresis in Practice: A Guide to Methods and Applications of DNA and Protein Separations* (5th ed.). Wiley-VCH.
6. Clinical and Laboratory Standards Institute (CLSI). Various guidelines on chromatography and electrophoresis methods.

ASSESSMENT QUESTIONS

1. Explain why ammonium sulfate precipitation is considered a "low-resolution" technique. When would you use it despite this limitation?
2. A protein has a pI of 6.5. Which type of ion-exchange chromatography would you use to purify it at pH 8.0? Explain your reasoning.
3. Compare and contrast SDS-PAGE and native PAGE. When would you choose one over the other?
4. Describe how you would separate a mixture containing albumin (66 kDa, pI 4.7), IgG (150 kDa, pI 7.0), and lysozyme (14 kDa, pI 11.0) using two different chromatographic techniques.
5. A clinical laboratory needs to measure vitamin D levels with high accuracy. Why would LC-MS/MS be preferred over immunoassay methods?
6. Explain the principle of isoelectric focusing and its clinical application in hemoglobinopathy diagnosis.
7. A patient presents with suspected maple syrup urine disease. Which separation technique(s) would you use for diagnosis, and what would you expect to find?

End of Lecture